

Stabilizing Kelly: Estimation Risk, Regime Conditioning, and Detection Boundary

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We attack hidden failure modes evinced in the Kelly-on-equities literature with a multi-layered regularization engine and evaluate them under realistic transaction costs on weekly data. A 2x2 factorial in covariance and mean shrinkage isolates the contribution of each pure shrinkage layer; two operational uses of a non-parametric macro regime label — one filtering the inputs to moment estimation, one scaling the resulting Kelly weights — isolate the data-side and sizing-side uses of regime information; and a stacked variant combines the strongest ingredients. Using 24 EuroStoxx 50 constituent equities over 2001–2018 with a 104-week rolling window and 10 bps one-side transaction costs, we find that mean-estimation noise — not covariance ill-conditioning — is the binding constraint on Kelly’s out-of-sample viability. The Σ -shrinkage main effect is essentially zero ($\Delta\text{Sharpe} = -0.003$); the μ -shrinkage main effect is small but real ($+0.06$); regime-conditional moment estimation is responsible for the bulk of the achievable improvement ($+0.40$), and the fully stacked Combined variant achieves a net Sharpe of 0.66 against 0.46 for long-only minimum variance and 0.09 for naive plug-in Kelly. A natural-experiment diagnostic shows that the rolling sample mean is statistically indistinguishable from zero for the median asset-window combination (median $|t| = 0.71$), explaining why static shrinkage produces only marginal gains while regime conditioning produces substantial ones. A formal delta-method Z-test on RegimeFilter vs Naive yields $Z = 1.60$, $p = 0.055$ at the boundary of detection; an analytical power calibration shows the 17-year panel under-powers the comparison but does not trivially fail it. A standalone Merton jump-diffusion Monte Carlo confirms that fractional Kelly is structural, not a concession to estimation noise: even with parameters known exactly, plugging them into the GBM Kelly formula reproduces the over-leverage pathology observed empirically.

Keywords: Kelly Criterion, Portfolio Optimization, Estimation Risk, Mean Shrinkage, Covariance Shrinkage, Regime Conditioning, Jorion Shrinkage, Jump-Diffusion, Sharpe Ratio Inference

1 Introduction

The Kelly Criterion, derived by Kelly (1956) in the context of information theory, provides a closed-form solution to the problem of optimal capital allocation under uncertainty. For a portfolio of n risky assets with expected return vector μ , covariance matrix Σ , and risk-free rate r , the criterion maximizes the expected logarithmic growth rate $g(\mathbf{f}) = r + \mathbf{f}^\top(\mu - r\mathbf{e}) - \frac{1}{2}\mathbf{f}^\top\Sigma\mathbf{f}$, yielding the unconstrained optimum $\mathbf{f}^* = \Sigma^{-1}(\mu - r\mathbf{e})$. Under known parameters this strategy is asymptotically growth-optimal (Breiman 1961; Thorp 2011). The optimum is also directly proportional to estimated excess returns and requires inversion of the covariance matrix, so even modest estimation errors in μ or Σ can produce allocations that are wildly aggressive, dangerously concentrated, or both.

Modern Kelly literature evinces a recurring set of obstacles that block this theoretical promise from translating to out-of-sample performance on equity panels. Naive plug-in implementations suffer from covariance ill-conditioning when the asset-to-observation ratio is non-negligible, from mean-estimation noise so severe that the rolling sample mean is statistically indistinguishable from zero for the typical asset-window, from regime contamination when historical moments pool observations across structurally different market environments, and from formula misspecification when the assumed return-generating process omits the fat-tailed jump dynamics that characterize real equity returns. These obstacles are well-documented across the empirical-optimization literature: Jorion (1986) and Michaud (1989) on mean-estimation error and “estimation-error maximization” in mean-variance optimization, DeMiguel, Garlappi, and Uppal (2009) on optimized portfolios losing to naive 1/N out-of-sample, and empirical investigations of long-only constrained Kelly on equity universes such as Carta and Conversano (2020) on concentration, drawdowns, and out-of-sample degradation. But there is a lack of systematic isolation in which obstacle is binding. The present study attacks each of these hidden obstacles with a uniquely-engineering regularization layer, evaluates them in isolation and in combination on a common rolling-window backtest with explicit transaction costs, and isolates which obstacle is binding.

The central finding is that plug-in Kelly’s well-known out-of-sample failure on equity panels is driven by μ -estimation noise — the rolling sample mean is statistically indistinguishable from zero for the median asset-window combination on a 104-week lookback — and that regularizing μ through *regime-conditional moment estimation* (rather than static shrinkage toward a fixed target) is sufficient to restore Kelly’s theoretical advantage. The fully stacked variant (**Combined**: μ -shrinkage and Σ -shrinkage applied to regime-filtered data) achieves an OOS Sharpe of 0.66 net of 10 bps transaction costs, against 0.46 for the long-only minimum-variance benchmark and 0.09 for naive plug-in Kelly. We further show that the same regime label operates very differently when used to filter inputs (**RegimeFilter**) versus to set a position-size multiplier (**RegimeDial**): the former optimizes for risk-adjusted return, the latter for drawdown control, and neither dominates. A standalone Merton jump-diffusion Monte Carlo confirms that the over-leverage pathology is structural even with parameters known exactly, completing the case that fractional scaling is a necessity rather than a concession to estimation noise.

1.1 Strategy lineup

Each Kelly variant uses identical optimization (long-only constrained QP, no leverage, no shorting) and differs only in how moments are estimated and how the resulting weight is scaled. The first four entries form a 2x2 factorial in Σ -shrinkage and μ -shrinkage, the next two isolate the data-side and sizing-side uses of the regime label, and the last stacks the strongest ingredients. The Naive

variant uses plug-in sample mean and sample covariance. SigmaShrink applies Ledoit and Wolf (2004) shrinkage on Σ while retaining the sample mean for μ , and MuShrink reverses the asymmetry by shrinking μ toward the period risk-free rate at $\alpha = 0.5$ while retaining the sample covariance; SigmaMuShrink applies both shrinkage layers jointly. RegimeFilter restricts moment estimation to historical observations matching the current regime and uses the full Kelly fraction, while RegimeDial estimates moments unconditionally but multiplies Kelly weights by a pre-registered regime-dependent fraction (1.0, 0.5, 0.5, 0.25) across (LowVolExp, HighVolExp, LowVolContraction, Crisis). Combined stacks SigmaShrink and MuShrink on regime-filtered data, pairing the strongest input-quality remedy with both shrinkage layers. Two benchmarks complete the comparison: MinVar, the long-only minimum-variance portfolio (depends only on Σ), and EqualWeight, the 1/N benchmark (depends on no estimation).

1.2 Emergent shrinkage in the computational pipeline

The named μ -shrinkage estimator (MuShrink) is only the most explicit shrinkage layer in this paper. Three further layers operate as μ -regularizers without being labeled as such, and we make this architecture visible because it determines how the empirical results should be read: the Sharpe gains are not the product of one designed-in shrinkage move applied to an otherwise naive pipeline, but of a stack of shrinkage-equivalent mechanisms, only one of which we introduce by name.

First, RegimeDial is mathematically identical to state-conditional Jorion shrinkage. Multiplying the long-only Kelly weights by a fraction ϕ produces exactly the QP solution that would obtain under the substitution $\tilde{\mu} = r\mathbf{1} + \phi(\hat{\mu} - r\mathbf{1})$, which is the convex combination $\tilde{\mu} = \alpha r\mathbf{1} + (1 - \alpha)\hat{\mu}$ with $\alpha = 1 - \phi$. The pre-registered dial values (1.0, 0.5, 0.5, 0.25) therefore correspond to shrinkage intensities $\alpha \in \{0, 0.5, 0.5, 0.75\}$ across the four regimes. The dial is not a “leverage rule” applied on top of the estimator; it is a regime-conditional μ -shrinkage estimator with intensity rising as the macro state deteriorates.

Second, the long-only constraint applies one-sided shrinkage on μ through the constraint set rather than through the estimator. When the unconstrained Kelly optimum demands a short position in asset i (i.e., when the sample mean for that asset falls below r), the long-only QP zeroes that component. The KKT-equivalent μ vector consistent with the constrained weights truncates sample-mean values below r at r itself: an asymmetric Jorion shrinkage applied only from below. This mechanism is responsible for a non-trivial fraction of the long-only constraint’s empirical advantage over unconstrained mean-variance optimization documented in Jagannathan and Ma (2003), and it operates silently in every variant we report.

Third, transaction costs function as a selection pressure in the cross-strategy comparison that favors estimators with implicit μ -stability. The TC-aware net Sharpe penalizes period-over-period weight churn, which in Kelly’s linear-in- μ regime translates directly to period-over-period μ -noise. Estimators that produce stable allocations — regime-conditional estimators, shrunk estimators — pay less of this penalty than naive estimators that chase the noisy pooled-sample mean.

The implication for reading the results: the +0.06 Sharpe gain attributed to MuShrink is the *visible* portion of μ -regularization in this pipeline. The +0.40 Sharpe gain attributed to RegimeFilter is partly conditional-mean estimation (a bias-variance improvement on the sample mean) and partly the same long-only and TC-driven mechanisms acting on a smaller, less noisy sample. The +0.36 Sharpe gain attributed to RegimeDial is, by the algebra above, *entirely* state-conditional Jorion shrinkage — but with α chosen by a non-parametric macro signal rather than by cross-validation.

None of the regularization in this paper is black-box: every shrinkage move is either explicit or expressible as an algebraic identity on the QP.

2 Theoretical Framework

2.1 The Kelly Criterion

For a single risky asset with expected return μ , volatility σ , and risk-free rate r , the growth-optimal allocation is $f^* = (\mu - r)/\sigma^2$. In the multi-asset setting the long-only Kelly portfolio maximizes $g(\mathbf{F}) = r + \mathbf{F}^\top(\mu - r\mathbf{1}) - \frac{1}{2}\mathbf{F}^\top\Sigma\mathbf{F}$ subject to $\sum_i f_i \leq 1$ and $f_i \geq 0$, the long-only no-leverage conditions standard in practical Kelly implementations on equity universes. This constrained quadratic program is solved via the Goldfarb-Idnani algorithm.

2.2 μ -Shrinkage toward the risk-free target

Static μ -shrinkage replaces the sample mean with the convex combination $\tilde{\mu} = \alpha \cdot \mu_{\text{target}} \cdot \mathbf{1} + (1 - \alpha) \cdot \hat{\mu}$, with $\alpha = 0.5$ and target equal to the period risk-free rate. This is a deliberately weak, model-free prior: it says nothing about cross-sectional structure or time variation, only that a noisy sample mean should be partially discounted. The empirical results in Section 4 show that this produces a modest but real Sharpe improvement (+0.06 over Naive); a stronger conditional prior — the regime-conditional historical mean used in RegimeFilter — captures roughly seven times that effect. Static μ -shrinkage is best understood as a baseline regularizer that earns its place in the headline **Combined** variant by adding additional Sharpe (+0.17) on top of regime filtering, where the within-regime sample is small enough that further regularization helps.

2.3 Σ -Shrinkage (Ledoit-Wolf)

The sample covariance matrix $S = X^\top X/n$ is ill-conditioned whenever p/n is non-negligible. Ledoit and Wolf (2004) show that sample eigenvalues are systematically more dispersed than their population counterparts. Their shrinkage estimator $\Sigma^* = \delta \cdot \mu_\Sigma I + (1 - \delta) \cdot S$ is the asymptotically optimal convex combination of S with the scaled identity matrix $\mu_\Sigma I$, where $\delta \in [0, 1]$ is determined analytically. At $p/n \approx 0.23$ on the weekly panel the sample covariance is not severely ill-conditioned, and the empirical LW intensity averages $\bar{\delta} \approx 0.08$ — which explains why the Σ -shrinkage main effect on Sharpe is essentially zero in our results.

2.4 Regime classification and conditioning

We classify each trading day into one of four macroeconomic regimes based on the equal-weighted portfolio’s rolling 60-day mean return $\bar{r}_t$ and rolling 60-day standard deviation σ_t , benchmarked against an *expanding* median of rolling volatility $\tilde{\sigma}_t$ (no look-ahead): Low-Vol Expansion ($\bar{r}_t > 0$, $\sigma_t < \tilde{\sigma}_t$), High-Vol Expansion ($\bar{r}_t > 0$, $\sigma_t \geq \tilde{\sigma}_t$), Low-Vol Contraction ($\bar{r}_t \leq 0$, $\sigma_t < \tilde{\sigma}_t$), and Crisis ($\bar{r}_t \leq 0$, $\sigma_t \geq \tilde{\sigma}_t$). The classifier is non-parametric: no HMM, no transition probabilities. The label is then deployed in two operationally distinct ways. *Data-side* (RegimeFilter): at each rebalancing date, moment estimation is restricted to historical observations classified into the same regime as

the current state, with a minimum of 30 observations required before falling back to the full sample. *Sizing-side* (RegimeDial): moments are estimated unconditionally but the resulting Kelly weights are multiplied by a pre-registered regime-dependent fraction (1.0, 0.5, 0.5, 0.25).

2.5 Jump-diffusion stress test

The empirical sections argue that fractional scaling is needed because of estimation noise. A standalone Monte Carlo under Merton’s (1976) jump-diffusion model lets us test whether fractional scaling is *also* needed under known parameters with fat tails. The dynamics under the physical measure are

$$d \log S_t = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t + J_t dN_t,$$

with $\lambda = 2$ jumps per year and jump sizes normally distributed with mean -0.02 and standard deviation 0.04 . This connects directly to the RegimeDial framing above: fractional-Kelly-under-fat-tails and state-conditional-Jorion-shrinkage are the same mathematical object reached from opposite directions.

3 Data and Methodology

3.1 Data

We construct a universe of 24 EuroStoxx 50 constituent equities with daily adjusted closing prices obtained from Yahoo Finance over January 2000 through December 2018. After requiring at least 90% non-missing coverage, forward-filling residual gaps, and removing assets with near-zero return variance, the final panel comprises 4,531 daily observations, 918 weekly observations (the primary backtesting series), and 211 monthly observations spanning 2001-05-24 to 2018-12-28. Weekly log returns are computed from end-of-week prices to ensure time-additivity consistent with the Kelly log-growth objective. The ratio $p/n = 24/104 \approx 0.231$ on the 104-week window confirms that the sample covariance matrix operates well below the boundary of ill-conditioning that would motivate large Σ -shrinkage intensities.

3.2 Backtesting framework

All strategies are evaluated through a common rolling-window out-of-sample backtesting framework operating on weekly log returns with a 104-week (two-year) lookback. At each rebalancing date, expected returns and the covariance matrix are estimated from the trailing window using the variant-specific estimator; portfolio weights are computed via constrained quadratic programming (long-only, no leverage) and applied to the subsequent week’s realized returns. Transaction costs are charged at 10 bps per unit of one-side turnover, with both gross and net returns recorded; the headline numbers reported throughout are net of transaction costs. Performance is evaluated using compound annual growth rate (CAGR), annualized volatility, Sharpe ratio, maximum drawdown, Calmar ratio, and Sortino ratio.

3.3 Statistical inference

We limit formal hypothesis testing to a single comparison — RegimeFilter vs Naive — and present the result alongside an analytical power calibration. The point is not to test every pairwise comparison (which would require multiplicity adjustments that obscure rather than clarify on a panel this size) but to demonstrate, with one well-chosen test, where the panel-length boundary actually lies. The test follows Lo (2002): assuming each strategy’s weekly return follows a stationary AR(1) with contemporaneously correlated innovations, the functional delta method applied to $g(\theta) = \mu_1/\sigma_1 - \mu_2/\sigma_2$ yields the asymptotic variance of $\widehat{SR}_1 - \widehat{SR}_2$ as a linear combination of innovation moments. We apply the test directly to the weekly OOS net-return series; weekly equity returns have weaker microstructure-driven autocorrelation than daily, making the diagonal VAR(1) approximation defensible without monthly aggregation, and a Ljung-Box check on AR(1) residuals confirms the working model.

4 Results

4.1 Out-of-sample performance

Table 1: Headline OOS Performance (Weekly, Net of 10 bps TC). All metrics computed on weekly returns over the OOS period using 104-week rolling estimation windows.

	CAGR	AnnVol	Sharpe	MaxDD	Calmar	Sortino	TerminalWealth
Naive	-0.0129	0.2865	0.0858	-0.8010	-0.0161	0.0918	0.8161
SigmaShrink	-0.0137	0.2866	0.0830	-0.8000	-0.0172	0.0886	0.8055
MuShrink	0.0096	0.2595	0.1429	-0.7589	0.0127	0.1561	1.1614
Sigma-MuShrink	0.0110	0.2587	0.1469	-0.7503	0.0147	0.1614	1.1874
RegimeFilter	0.1023	0.2356	0.4899	-0.6839	0.1496	0.6337	4.5946
RegimeDial	0.0788	0.1844	0.4494	-0.4233	0.1861	0.6019	3.2763
Combined	0.1377	0.2174	0.6570	-0.6040	0.2280	0.8646	7.5322
MinVar	0.0762	0.1722	0.4575	-0.6129	0.1243	0.5334	3.1569
EqualWeight	0.0607	0.1980	0.3488	-0.5619	0.1081	0.4357	2.5169

First, the 2x2 shrinkage factorial isolates the contribution of each pure regularization layer. The Σ -shrinkage main effect is essentially zero (SigmaShrink Sharpe 0.083 vs Naive 0.086 net of TC; SigmaMuShrink 0.147 vs MuShrink 0.143). This is consistent with the LW intensity diagnostic ($\bar{\delta} \approx 0.08$): at $p/n \approx 0.23$ the sample covariance is not severely ill-conditioned, so there is little for shrinkage to repair. The μ -shrinkage main effect is small but real (+0.06 Sharpe across both Σ -conditions), and the interaction is approximately additive.

Second, regime conditioning is responsible for the bulk of the achievable improvement. RegimeFilter alone (+0.40 Sharpe over Naive) outperforms every static-shrinkage variant by roughly an order of magnitude, and exceeds MinVar’s risk-adjusted return (Sharpe 0.49 vs 0.46) despite carrying meaningfully higher volatility. The mechanism is implicit: regime-conditional moment estimation is a strong prior on μ , in contrast to a weak fixed-target prior that static MuShrink provides.

Third, the two operational uses of the regime label are complementary. RegimeFilter (data-side) maximizes Sharpe but accepts a -68% peak-to-trough drawdown; RegimeDial (sizing-side) accepts a 0.04 lower Sharpe in exchange for shrinking maximum drawdown to -42% . Their Calmar ratios (0.15 and 0.19 respectively) reverse the Sharpe ranking, indicating that the right choice depends on whether the investor optimizes for risk-adjusted return or for return-per-unit-of-pain. Filter and Dial are best read as a Pareto frontier of regime-information uses: a Sharpe-maximizing mandate selects Filter, a drawdown-minimizing mandate selects Dial, and a mandate balancing both selects Combined. The fully stacked Combined variant — shrinkage layers applied to regime-filtered data — inherits Filter’s drawdown profile but extracts an additional $+0.17$ Sharpe (0.66 net). The within-regime sample is small enough that further shrinkage is no longer redundant.

Fourth, Combined’s terminal-wealth advantage is earned in the recovery, not the crisis. Figure 1 shows that all full-Kelly variants (including Combined) suffered drawdowns above -60% in the 2008-2009 unwind, comparable to MinVar’s -61% . Combined separates from the rest through the 2010-2018 recovery, where the regime classifier increasingly populated the Low-Vol Expansion state and the within-regime moment estimates aligned with realized returns. RegimeDial is the only variant in this universe that materially contained the 2008 drawdown.

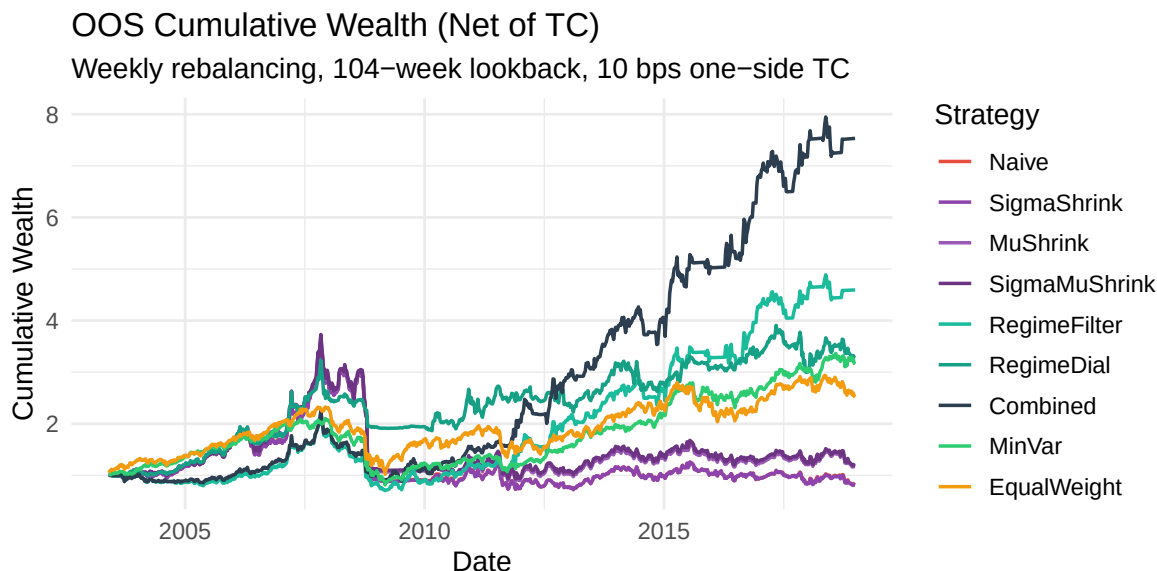


Figure 1: OOS cumulative wealth paths, net of 10 bps one-side TC.

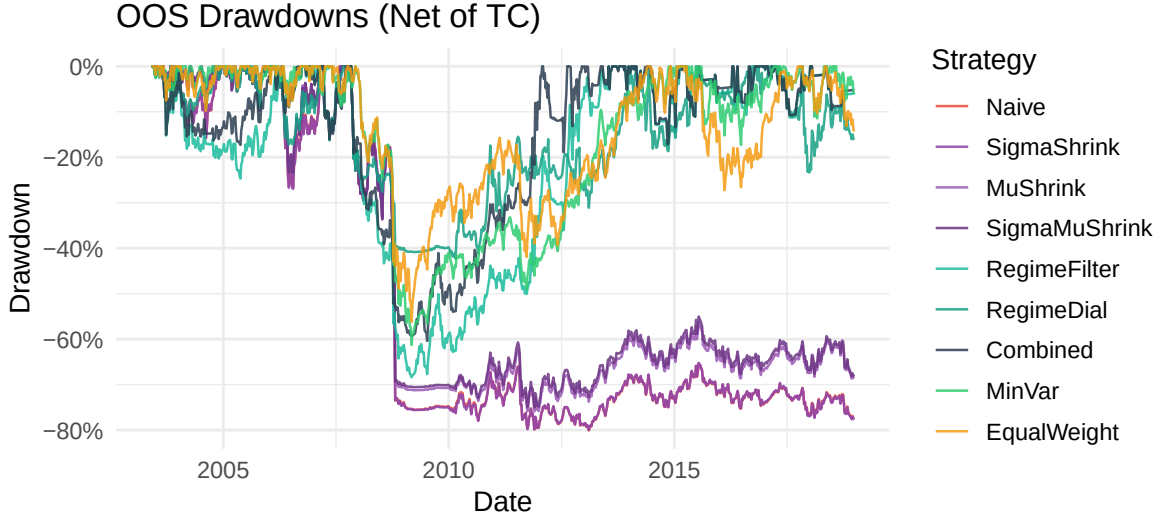


Figure 2: OOS drawdown profiles, net of TC.

4.2 Turnover and transaction-cost wedge

Table 2: Turnover Diagnostics by Kelly Strategy (one-side units). Mean / Median / Max are per-period turnover. Ann. Turn (annualized) times the 10 bps TC rate gives the gross-vs-net Sharpe wedge. Mean N / Median N are the active position counts.

	Mean	Median	Max	Ann. Turn	Mean N	Median N
Naive	0.258	0.081	1.707	13.423	1.509	1
SigmaShrink	0.257	0.107	1.680	13.386	1.550	1
MuShrink	0.245	0.174	1.558	12.738	1.969	2
SigmaMuShrink	0.244	0.186	1.392	12.671	2.053	2
RegimeFilter	0.358	0.121	2.000	18.595	1.525	2
RegimeDial	0.229	0.107	1.707	11.929	1.509	1
Combined	0.321	0.095	2.000	16.675	1.982	2

Annualized turnover ranges from 11.9 (RegimeDial) to 18.6 (RegimeFilter) for the Kelly variants, against 4.1 for MinVar and 0.06 for EqualWeight. Multiplying by the 10 bps per-unit cost gives Sharpe wedges between 12 and 19 bps annualized for the Kelly strategies — well below the levels at which transaction costs would dominate returns. The naive plug-in pipeline has the second-lowest median turnover (0.081) but the highest mean (0.258), reflecting its tendency to oscillate between concentrated single-name allocations as the noisy sample mean shifts. RegimeFilter’s higher turnover is partly mechanical: regime transitions trigger discrete shifts in the historical conditioning set, which in turn shift the conditional mean estimate.

4.3 Mean-variance efficient frontier

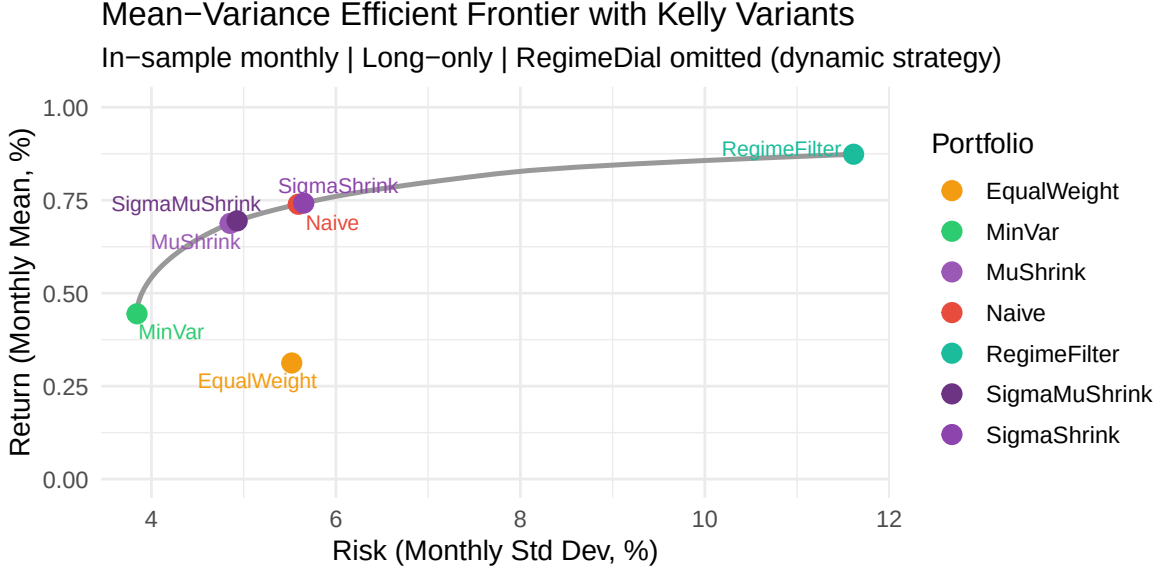


Figure 3: Mean-variance efficient frontier (grey) with Kelly variants and benchmarks; in-sample monthly moments, long-only, RegimeDial omitted.

The frontier provides a useful visual diagnostic: portfolios lying on or near the frontier are mean-variance efficient given the in-sample moments, while distance from the frontier reflects the cost of estimation error or suboptimal allocation. RegimeFilter sits at the upper-right extreme, consistent with its conditional moment estimator targeting higher-return assets in low-vol-expansion regimes; the Naive / SigmaShrink / MuShrink / SigmaMuShrink cluster sits in the tangency neighborhood of the frontier, with Σ -shrinkage and μ -shrinkage producing minor coordinate shifts; MinVar anchors the leftmost point; EqualWeight lies inside the frontier, as expected for an estimation-free heuristic. The key insight is that all Kelly variants target positions on or near the in-sample frontier but diverge substantially from these positions out of sample — a visual confirmation that estimation risk, not allocation methodology, is the dominant source of performance degradation. RegimeDial is omitted because its dynamic regime-dependent leverage produces a different point each period; a single static visualization would mislead.

4.4 Natural experiment: signal-to-noise ratio of the sample mean

Plug-in Kelly's failure on the static and pure-shrinkage variants is driven by μ -estimation noise on the rolling windows the optimizer is forced to consume. To quantify that noise directly, for each asset i and rolling window of T weekly observations we compute the standard error of the sample mean $\hat{\sigma}_i/\sqrt{T}$ and the t-statistic $t_i = \hat{\mu}_i/(\hat{\sigma}_i/\sqrt{T})$. Under a null of zero true mean, t_i is asymptotically standard normal, so $|t_i| < 1.96$ indicates the sample mean is not significantly different from zero at conventional levels. Pooled across all 24 assets and 814 rolling windows ($n = 19,536$ t-statistics), the median $|t|$ is 0.71, and only 7.2% of windows have $|t| > 1.96$ against a null expectation of 5.0%.

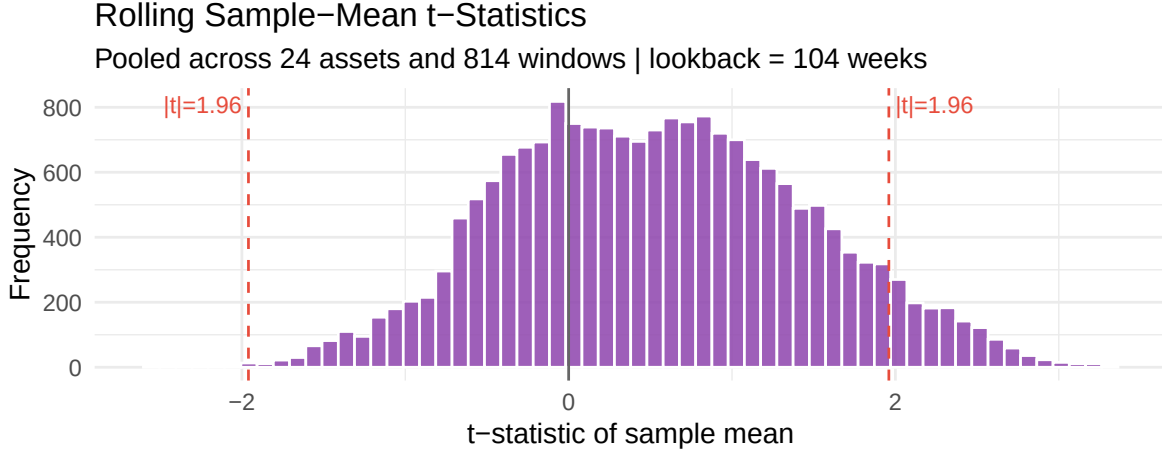


Figure 4: Pooled rolling-window sample-mean t-statistics across 24 assets and 814 windows; 104-week lookback.

The interpretation: the rolling sample mean fed to plug-in Kelly is, for the median asset-window combination, statistically indistinguishable from zero. The 1.4x excess of windows above the 95% threshold confirms that genuine cross-sectional signal exists but is sparse and asset-specific. This explains the empirical pattern in Table 1: static μ -shrinkage toward r produces a small Sharpe gain (+0.06) because it correctly discounts noise but cannot exploit the sparse signal; regime-conditional moment estimation produces a much larger gain (+0.40) because conditioning on the current macroeconomic state captures structure that pooled-sample estimation washes out.

4.5 Regime diagnostics

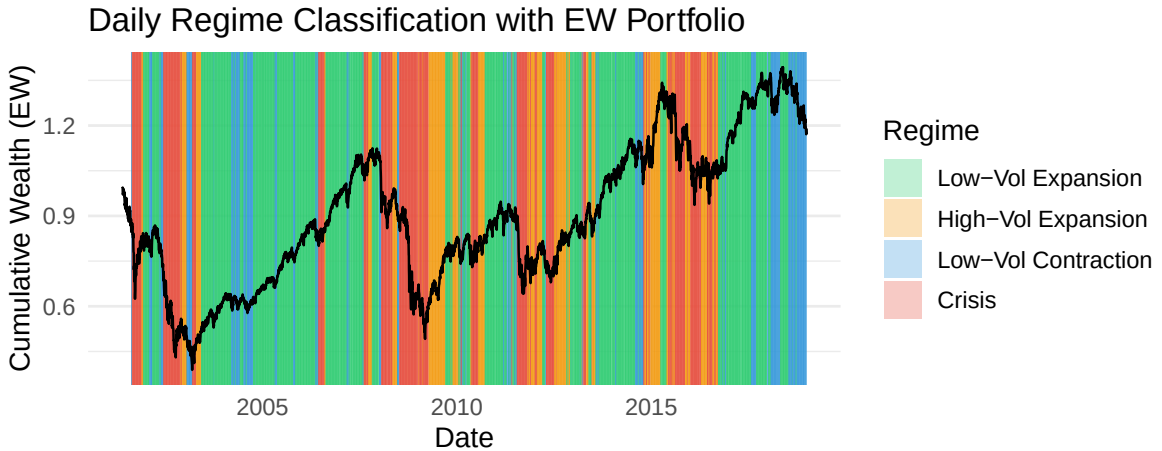


Figure 5: Daily regime classification overlaid on the equal-weighted portfolio's cumulative wealth.

The mean Ledoit-Wolf shrinkage intensity over the rolling 104-week windows is $\bar{\delta} = 0.083$, with a single spike to $\delta \approx 0.36$ during the 2008-2009 financial crisis that quickly subsides. This confirms

that on the weekly $p/n \approx 0.23$ panel the sample covariance is well-conditioned by construction and is the mechanical reason for the absence of any Σ -shrinkage main effect on Sharpe.

4.6 Statistical inference at the detection boundary

Table 3: Delta-method Z-test: $\text{SR}(\text{RegimeFilter}, \text{net}) > \text{SR}(\text{Naive}, \text{net})$, weekly. Ljung-Box diagnostics report p-values for residual autocorrelation in the AR(1) innovations; values above 0.05 indicate residuals are consistent with the diagonal VAR(1) working model.

Quantity	Value
T (weeks)	814
Sharpe (RegimeFilter, ann.)	+0.4902
Sharpe (Naive, ann.)	+0.0859
SR diff (annualised)	+0.4044
SE(SR diff)	0.0351
Z statistic	+1.5959
p-value (one-sided)	0.0553
ρ (RegimeFilter)	-0.1064
ρ (Naive)	-0.0190
innovation correlation	+0.4655
Ljung-Box p (RegimeFilter)	0.1218
Ljung-Box p (Naive)	0.5650

The test puts the paper at the boundary of detection, not beyond it: with $T = 814$ weekly observations the test returns $Z = 1.60$, $p = 0.055$ (one-sided) on an annualized Sharpe difference of 0.40. The Ljung-Box diagnostics on the AR(1) residuals are both well above 0.05, indicating the diagonal VAR(1) approximation is not detectably misspecified.

The Z-test returns a per-observation variance V such that $\text{Var}[\widehat{SR}_1 - \widehat{SR}_2] = V/T$. We use this to compute the analytical power of the test as a function of sample size, holding V at its empirical value: $\text{power}(T, \Delta, \alpha) = 1 - \Phi(z_{1-\alpha} - \Delta\sqrt{T/V})$.

Table 4: Sample size required for 80% power, given $V = 1.005$.

Delta SR (weekly)	Delta SR (annualised)	T80 (weeks)	T80 (years)
0.0236	0.17	11180	215.0
0.0485	0.35	2637	50.7
0.0957	0.69	679	13.1

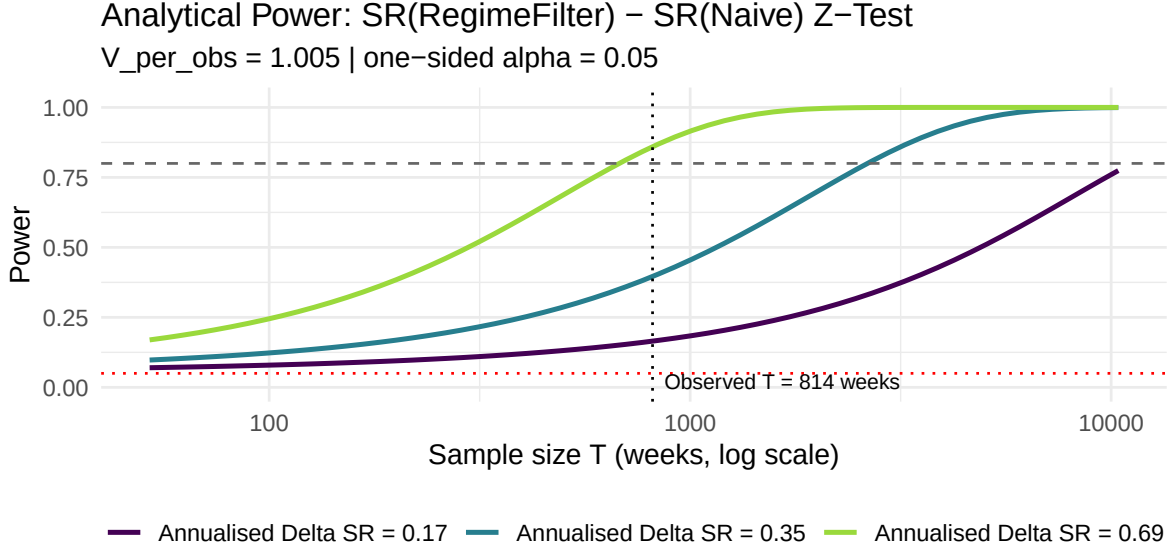


Figure 6: Analytical power for the SR(RegimeFilter) - SR(Naive) Z-test by sample size, at three annualized Δ SR effect sizes.

The table reads as follows: under the empirical V , detecting a true annualized Sharpe difference of 0.17 at 80% power requires roughly 215 years of weekly data, of 0.35 requires 51 years, of 0.69 requires 13 years. The observed RegimeFilter-vs-Naive annualized difference of 0.40 sits between the second and third grid rows. Two readings are honest. *The effect is large and the test is nearly significant.* If the true Sharpe gap is the observed 0.40, a panel of roughly 30 years would place the test comfortably below conventional thresholds; the 17-year panel under-powers the comparison but does not trivially fail it. *Smaller hypothesised effects — for example, the +0.06 Sharpe gain from static μ -shrinkage — are categorically undetectable on any panel a single research project would assemble.* This makes the broader testing posture clear: pairwise tests across all variant comparisons would mostly fail to reject not because the variants are equivalent but because the panel cannot resolve them.

4.7 Jump-diffusion stress test

Table 5: GBM vs Jump-Diffusion Terminal Wealth (~20-year MC). P_Loss is the probability of ending below initial capital; P_Ruin the probability of ending below 1 percent of initial; P_W10 the probability of exceeding 10x initial capital.

Strategy	Model	Median	P_Loss	P_Ruin	P_W10
Half	GBM	2.1781	0.0980	0.0000	0.0045
Full	GBM	2.5769	0.2190	0.0000	0.1320
Double	GBM	1.0655	0.4900	0.0250	0.1785
Triple	GBM	0.1369	0.7090	0.2340	0.1120
Half	JumpDiffusion	1.6971	0.2045	0.0000	0.0015
Full	JumpDiffusion	1.3491	0.3975	0.0000	0.0700

Strategy	Model	Median	P_Loss	P_Ruin	P_W10
Double	JumpDiffusion	0.4234	0.6405	0.0655	0.1015
Triple	JumpDiffusion	0.0244	0.8395	0.4080	0.0570

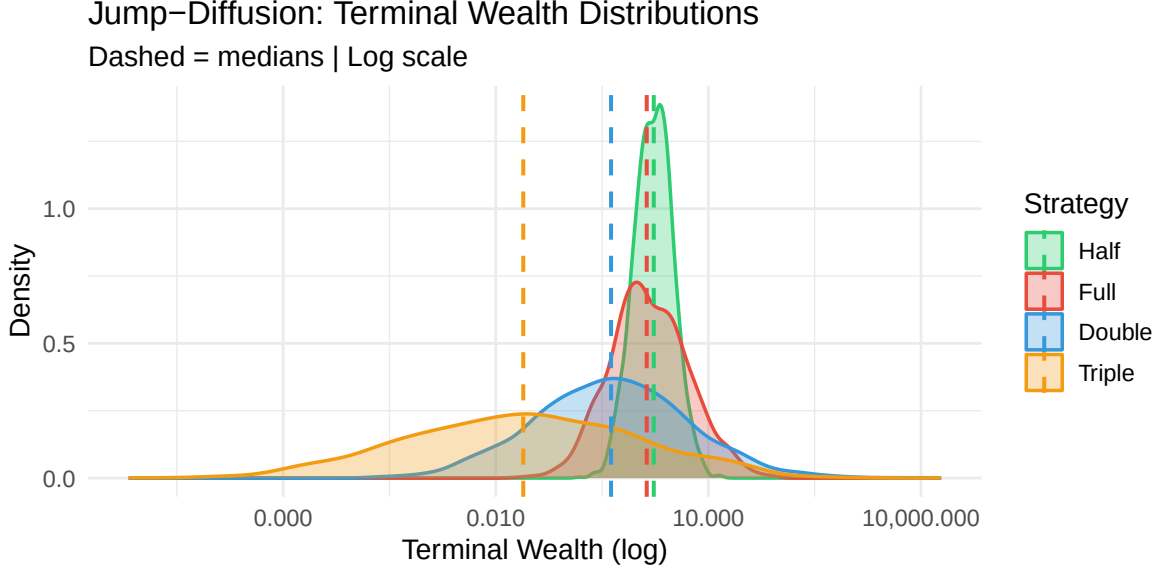


Figure 7: Terminal wealth distributions under Merton jump-diffusion across four Kelly fractions; dashed lines mark medians.

We deliberately apply the textbook GBM Kelly formula $f^* = (\mu - r)/\sigma^2 = 0.6875$ without correcting for the jump compensator $\lambda \cdot \mathbb{E}[e^J - 1] \approx -0.038$ per year. This is the realistic case: a practitioner who has correctly identified μ , σ , λ , and the jump distribution but reaches for the closed-form Kelly fraction will plug them into the GBM formula. The numerical growth-optimum under the actual DGP — found by maximizing $\mathbb{E}[\log W_T]$ on simulated paths — is approximately $f_{\text{true}}^* \approx 0.38$, so the “Full Kelly” line is in fact $\approx 1.8 \times$ true Kelly, and the “Half Kelly” line at $0.5 \times 0.6875 = 0.34$ lands close to the actual optimum. The exercise demonstrates that even with all model parameters known exactly, plugging them into a misspecified Kelly formula reproduces the same over-leverage pathology observed in the empirical sections under estimation noise — fractional scaling is structural, not a concession to noise.

5 Discussion

The results reveal a clear hierarchy among the regularization layers and illuminate the structural reasons for their differential effectiveness.

The binding constraint is μ -regularization, not Σ -regularization. The 2x2 shrinkage factorial provides a clean isolation of main effects: at $p/n \approx 0.23$ on the weekly panel, with mean Ledoit-Wolf intensity $\bar{\delta} \approx 0.08$, the sample covariance is well-conditioned and Σ -shrinkage produces no detectable Sharpe improvement. The complementary diagnostic — median $|t| = 0.71$ on the

rolling 104-week sample mean, with only 7.2% of windows above the 95% threshold against a 5% null expectation — documents directly why μ -regularization matters more: the sample mean fed to the optimizer is, for the typical asset-window combination, statistically indistinguishable from zero. Static μ -shrinkage toward r extracts the small Sharpe gain (+0.06) consistent with discounting noise, but cannot exploit the sparse asset-specific signal the t-statistic excess hints at.

Regime conditioning works because it addresses the right problem. RegimeFilter’s +0.40 Sharpe gain is roughly seven times the static μ -shrinkage effect because the regime-conditional historical mean is a much stronger prior than a fixed risk-free target. By restricting parameter estimation to observations from the current macroeconomic state, the estimator captures structure that pooled-sample estimation washes out: crisis-period correlations, expansion-period drift, and the regime-dependent volatility levels that drive the regime classifier in the first place. The 2001-2018 sample spans multiple distinct regimes — the dot-com aftermath, the 2003-2007 bull market, the 2008 financial crisis, the European sovereign debt crisis, and the 2012-2018 recovery — and conditioning on the current state rather than pooling across them is the single largest improvement.

Filter and Dial are complementary, not competing. The two operational uses of the regime label trade Sharpe against drawdown. RegimeFilter’s −68% peak-to-trough drawdown and 0.49 Sharpe versus RegimeDial’s −42% drawdown and 0.45 Sharpe expose a Pareto frontier of regime-information uses: a Sharpe-maximizing mandate selects Filter, a drawdown-minimizing mandate selects Dial. The Calmar ratio reverses the Sharpe ranking (0.19 for Dial vs 0.15 for Filter), reinforcing that the choice depends on the loss function. The Combined variant inherits Filter’s drawdown profile and adds shrinkage layers that earn an additional +0.17 Sharpe; this is the headline weekly net Sharpe of 0.66.

The shrinkage architecture is more layered than it appears. The named MuShrink estimator is only the most visible μ -regularizer in the pipeline. RegimeDial is mathematically identical to state-conditional Jorion shrinkage with intensities $\alpha \in \{0, 0.5, 0.5, 0.75\}$; the long-only constraint applies one-sided shrinkage by truncating sample means below r at r itself; transaction costs in evaluation function as a selection pressure favoring estimators with implicit μ -stability. None of this is hidden machinery, but merits focus as the empirical results should be read as the cumulative effect of a stack of shrinkage-equivalent mechanisms, not a single designed-in shrinkage move.

Statistical inference puts the paper at the boundary of detection. The single delta-method Z-test we run (RegimeFilter vs Naive) returns $Z = 1.60$, $p = 0.055$ on an annualized Sharpe difference of 0.40 — above the natural reading of the data but below conventional thresholds. The analytical power calibration makes the tradeoff explicit: detecting a 0.17 annualized Sharpe difference at 80% power requires 215 years of weekly data; detecting 0.35 requires 51 years; detecting 0.69 requires 13 years. The 17-year panel under-powers the headline comparison but does not trivially fail it, and we do not run pairwise tests for the smaller Sharpe differences because they are categorically undetectable on any panel a single research project would assemble. The honest reading is that the economic effect is large and the statistical evidence is suggestive but not decisive.

Jump-diffusion results reinforce the case for fractional Kelly. Even under known parameters, applying the GBM Kelly formula to a jump-diffusion DGP reproduces the over-leverage pathology observed empirically under estimation noise. The numerical growth-optimum under the true DGP is approximately $f_{\text{true}}^* \approx 0.38$, so what we label “Full Kelly” is in fact $\approx 1.8\times$ the actual optimum. This converges from a different direction with the empirical RegimeDial story: state-conditional Jorion shrinkage and fractional-Kelly-under-fat-tails are the same mathematical object, and both arguments prescribe the same operational rule — size positions below the textbook formula’s optimum.

6 Conclusion

We set out to determine which of the obstacles we distilled from contemporary Kelly-on-equities literature — covariance ill-conditioning, mean-estimation noise, regime contamination, and formula misspecification under fat tails — is the bottleneck on out-of-sample performance, and which targeted regularization layer is sufficient to defuse it. The answer is structured. The Σ -shrinkage main effect is essentially zero on the well-conditioned weekly panel; the μ -shrinkage main effect is small but real; regime-conditional moment estimation is responsible for the bulk of the achievable improvement, and the fully stacked Combined variant achieves a net Sharpe of 0.66 against 0.46 for long-only minimum variance and 0.09 for naive plug-in Kelly — a gap large enough to be economically meaningful and at the boundary of statistical detection on the 17-year panel. The two operational uses of regime label expose a Pareto frontier between Sharpe and drawdown, with choice depending on loss function.

Three practical conclusions follow. First, the binding constraint on Kelly’s out-of-sample viability is mean estimation noise, not covariance instability — a conclusion supported both by the natural-experiment t-statistic diagnostic and by the empirical hierarchy of regularization effects. Second, regime conditioning is the most promising single avenue for Kelly stabilization on equity panels because it operates as a strong, data-driven prior on μ rather than a weak fixed-target shrinkage. Third, jump-diffusion dynamics destroy the optimality of the GBM Kelly formula even under known parameters, making fractional scaling a structural necessity rather than a practical convenience. Future work should explore Bayesian return forecasting with macro factor priors, cross-validated regime-conditional shrinkage intensities, and explicit jump-aware Kelly formulas as paths toward closing the rift between Kelly’s theoretical and empirical performance.

References

- Breiman, L. (1961). Optimal gambling systems for favorable games. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, 1, 65–78.
- Carta, A., & Conversano, C. (2020). Practical implementation of the Kelly Criterion. *Frontiers in Applied Mathematics and Statistics*, 6, 577050.
- DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *Review of Financial Studies*, 22(5), 1915–1953.
- Jagannathan, R., & Ma, T. (2003). Risk reduction in large portfolios: Why imposing the wrong constraints helps. *Journal of Finance*, 58(4), 1651–1683.
- Jorion, P. (1986). Bayes-Stein estimation for portfolio analysis. *Journal of Financial and Quantitative Analysis*, 21(3), 279–292.
- Kelly, J. L. (1956). A new interpretation of information rate. *Bell System Technical Journal*, 35(4), 917–926.
- Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- Lo, A. W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Merton, R. C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2), 125–144.
- Michaud, R. O. (1989). The Markowitz optimization enigma: Is “optimized” optimal? *Financial Analysts Journal*, 45(1), 31–42.
- Thorp, E. O. (2011). The Kelly criterion in blackjack, sports betting, and the stock market. In L. C. MacLean, E. O. Thorp, & W. T. Ziemba (Eds.), *The Kelly Capital Growth Investment Criterion* (pp. 789–832). World Scientific.